

Translating clinical narratives into structured evidence

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Translating clinical narratives into structured evidence with neuro-symbolic systems

Abstract—Clinical information extraction (IE) converts clinical narratives into structured facts. Large language models can do this with significantly less preprocessing than rules-based methods, but they introduce new classes of problems: hallucination, schema drift and overconfidence. We propose a hybrid system where the model extracts entities, interprets relations and provides evidence, and the rules normalise entities, verify evidence and apply policies to resolve conflicts. This modular system preserves the rich uncertainty captured in clinical narratives while producing auditable structured outputs that confirm to explicit rules. We use 2 public datasets of epilepsy letters to evaluate this system on two distinct tasks: (1) selecting the most important clinical fact among many and (2) building a broad inventory of clinical facts. We show that the combined system improved F1 by +0.05 vs. rules-only or LLM-only systems, while creating a rich evidence trail with an exact evidence score of ≥ 0.99 . This transparent, evidence-grounded system is a step towards making more trustworthy systems for clinical use.

Index Terms—information extraction, neuro-symbolic, evidence verification, natural language processing

I. INTRODUCTION

Clinical information extraction is the process of turning clinical narratives into structured facts suitable for clinical use and research.

A. Medical framing

Epilepsy is one of the most common neurological diseases. 30 percent of patients are drug resistant and research progress has been stubbornly slow. We need a deeper understanding of who these patients are, how they are treated, how outcomes differ over time and across groups. The most important clinical markers are trapped in unstructured notes. Costly chart review is required for constructing patient timelines, identifying research cohorts, conducting retrospective studies.

They struggle through that anyway because it's so valuable, but it's not just costly it's error prone.

Entity extraction vs. entity relational extraction and reasoning. NLP solved entity extraction a while back, but this is a temporal and clinical reasoning problem - how to represent a clinical state in a rigorous way. It's not just computers that struggle with this problem, but doctors too - they all describe things differently and experts struggle to interpret them identically

B. Prior system summary

Rules-based systems are good at entity extraction but require extensive annotation and perform poorly on logic. BERT-based systems require extensive annotations and again, perform poorly on logic. GPT-based systems are very semantically

flexible and strong on reasoning, but they're a bit too flexible - hallucination, rambling prose, it's unstructured data to unstructured data. Monolithic LLMs provide the least visibility into the most dangerous task, clinical reasoning.

Most of the LLMs tested and performing well as monolithic systems were closed-source, which creates privacy concerns. These problems lead to systems generalising poorly to new tasks and styles. We need the modularity and reliability of rules-based systems to constrain the flexibility of the LLMs and audit their outputs.

C. Hybrid solution

I present a modular, evidence-grounded LLM framework for clinical information extraction from epilepsy clinical letters. It decomposes extraction into configurable stages for semantic interpretation, relation extraction, temporal resolution, concept normalisation, schema validation and evidence verification. It is evaluated across two independently developed, publicly shareable epilepsy benchmarks assessing two different capabilities: broad phenotype extraction in ExECTv2 and temporally complex seizure-frequency extraction from synthetic letters.

By directly comparing rules-only, LLM-only and hybrid configurations, plus component ablations and error analysis, we show how modular, evidence-grounded hybrid systems not only improve field-level F1 scores but also reliability, auditability and generalisability.

II. LITERATURE REVIEW

Epilepsy clinic letters contain diagnoses, seizure frequency, medication, investigations, and treatment changes, often distributed across time, hedged, and tied to more than one event. Several of those statements can be true in the same letter. Published extractors have already chosen which of them to keep. Fonferko-Shadrach *et al.* treat the letter as an inventory of supported facts across diagnosis, frequency, prescriptions, and investigations [1], [2]. Xie *et al.* [3], Holgate *et al.* [4], and Gan *et al.* [5] treat it as one current seizure-frequency state. These are different questions, and a score on one gold does not answer the other. Structured extraction is therefore not only recognition of medical terms. It is a decision about which true statements the evaluation form is allowed to keep [6].

Clinical information extraction, as used here, fills a pre-defined form; it is not full understanding of the letter. By the Message Understanding Conferences the task was already template fill against a prepared answer key [7]. Wang *et al.* define clinical IE as the automatic extraction and encoding of clinical information from text [8]. Fu *et al.* separate mention

detection from concept encoding: a quoted span is not the coded concept [6]. Later mapping into the evaluation form is therefore a named step, and it is older than large language models. Xie *et al.* extract a frequency sentence and then apply rules to obtain a number [9]. Agrawal *et al.* define a resolver that maps model text onto the evaluation form and report the model with and without that stage [10]. In much of the remaining epilepsy literature that later mapping is unpublished, and the headline score is the already-collapsed object: a band, a monthly number, or a feature-identical set. Rule steps that only write the form are also often unnamed [11]. The distinction required here is between encode, which writes the already-chosen fact in the designed string, and select, which may change the fact.

Most published systems are trained or tuned on one local corpus. Decker *et al.* report seizure-frequency F_1 of 0.82 in-house and 0.40 at a second centre; a rule stack that fits one letter style need not transfer [?]. That result is a caution about local assumptions, not an argument against recorded rules. Hybrids that mix models with rules are also not new, and they typically headline one number. The present work uses two public golds that already encode the two questions, names who runs extract, encode, and select (rules, a model, or both), and reports the select stop. Extract and encode are prior-stage ablations. The recorded object keeps the source span and the hop, not only the score.

III. METHODS

A. Data

Two datasets were used to evaluate the performance of the system on two distinct tasks. Dataset 1 assesses the system’s ability to extract a broad inventory of clinical facts and attributes, often with multiple facts per category e.g. multiple kinds of medication, each with its own dose, unit and frequency. Dataset 2 assesses the system’s ability to extract a single clinical fact that represents the patient’s current seizure frequency burden. This often involves multiple seizure frequency events that need to be aggregated (e.g. 1 in February, 2 in April = 3 per 3 months) or disambiguated (e.g. ”She discontinued Valproate on 10 Jul. Shortly afterwards, she experienced 2 to 3 seizures, one triggered by missed medication. She has remained seizure-free since”) to produce a single answer.

One dataset is tasked with selecting a single fact, the patient’s current seizure burden, from a 400-word letter. The other is tasked with selecting selecting an average of six facts per 200-word letter, covering a broad inventory of clinical facts (medication, diagnosis, investigations, seizure frequency).

Dataset 1 consists of 200 synthetic letters written by trained specialists based on real clinic letters from hospital outpatient consultations with epilepsy specialists in Swansea Bay. The letters were double annotated by four trained researchers and clinicians according to strict annotation guidelines. The dataset consists of 9 distinct categories describing a patient’s experience of epilepsy. In this paper we focused in the four most important categories: the medication used (drug name,

dose, unit, frequency), the diagnosis (type of seizures e.g. tonic-clonic seizures), the investigations used to confirm the diagnosis (e.g. MRI results) and how often the patient has seizures (e.g. focal seizures for around 1 year before this). An example letter is shown in Figure 1a. The dataset was split into

Dataset 2 consists of 1,500 letters using a combination of human and generative AI. Ten synthetic letter templates were created based on real clinic letters from King’s College Hospital. GPT-5 generated 500 diverse short descriptions of seizure frequency. Human annotators read and assigned gold-standard labels to each description. The description-label pairs were then inserted into the parameterised templates to create the full dataset of 15,000 letters. This paper used a subset of 1,500 letters provided by the author. The gold-standard annotation consists of a single label describing the patient’s current seizure frequency burden, e.g. seizure free for 5 months. An example letter is shown in Figure 2a.

Dear Dr,

Re: Mr Harry Harris. DOB 31/02/1950

Diagnosis: Symptomatic structural epilepsy
Right MCA infarct 2005

Seizure type and frequency: Focal motor seizures, (left arm jerks) 2-3 per month
Focal to bilateral convulsive seizures, last event 2015

Investigations: CT Head 1/3/2015 gliosis consistent with previous R MCA territory infarct

Medication: Carbamazepine 400mg bd
Sodium Valproate 500mg bd
Clopidogrel 75mg od
Simvastatin 20mg od
Ramipiril 5mg od

I reviewed this 67-year-old man by telephone today. His epilepsy was diagnosed in 2015 after he had a convulsive epileptic seizure for the first time. With hindsight he had been having focal seizures, consisting of left arm jerks, for around 1 year before this. The focal seizures were occurring more frequently, perhaps once per day before the carbamazepine was introduced alongside the sodium valproate. He has previously tried levetiracetam but that caused significant mood change.

As well as his cerebrovascular disease, he has high blood pressure and also gets occasional migraine. Interestingly his brother also has had a stroke and has developed post-stroke epilepsy, there is no other relevant family history. He was born prematurely at 32 weeks and weighed 2kg but had no developmental problems. He had 2 febrile seizures at the age of 8 months and 18 months. He has not had meningitis or encephalitis but has a significant head injury whilst playing rugby where he was unconscious for 1 hour and was kept in hospital overnight.

He was working as a surveyor before the stroke but has stopped working since. He needs a stick to walk. He lives with his wife and has stopped smoking, previously having smoked cigars for around 35 years. He drinks 8 pints of dark stout a week.

Mr Harris would like to get better control of his epilepsy - the focal seizures are troublesome and so in the first instance I would suggest increasing the carbamazepine to 600mg bd. I will review him again in clinic and arrange epilepsy nurse follow up.

KINGS NEUROSCIENCES CENTRE

Clinic Date: 10 June 2023

Dr Wang
Saffron Park Hospital

Saffron Park, London, E14 7JL

Dear Dr Wang

John Doe, DOB: 21-11-1982, Hospital No: P546484 NHS No. 5484656746
Flat 15 Roundwood Road, London, E14 7JL

I reviewed the above patient in the Neurology Clinic today. Since our last contact, he has adopted a pillbox with phone reminders and reports that this has markedly reduced missed doses and improved his day-to-day stability. He feels more confident managing his medications and describes better energy levels with fewer fluctuations during the week.

On Carbamazepine monotherapy he was seizure-free for 5 months, until a focal impaired-awareness seizure occurred three Thursdays ago. He recovered without injury and did not require hospital attendance. There were no witnessed convulsions, tongue biting or incontinence, and he reports an aura of brief disorientation followed by several minutes of confusion. He attributes the breakthrough event to a particularly disrupted week of sleep and work stress, though he did not miss any medication doses. He remains otherwise well and continues to use the adherence aids reliably.

Examination today is unremarkable and there are no new adverse effects reported beyond occasional mild tiredness on late clinic days, which he feels is manageable. He denies rash, pruritus or mood change. We discussed ongoing self-management, the importance of sleep regularly, and continuing the current adherence strategy. Given the overall improvement, the prolonged seizure-free interval, and the isolated breakthrough event, I have not made any changes to his current Carbamazepine regimen today. We agreed to continue observation with the present plan and to review if there are any further events.

I will follow up in three months, or sooner should he experience additional seizures or any concerns.

Yours sincerely

(a) Example clinical letter from the Swansea Bay with relevant facts about medication, diagnosis, investigations and seizure frequency highlighted.

(b) Example clinical letter from King's College Hospital with relevant seizure frequency events highlighted.

Fig. 1: Example clinical letters from Swansea Bay (a) and King's College Hospital (b). Highlighted sections of the letter illustrate the areas where the pipeline identified relevant clinical facts for the task.

B. Architecture

IV. RESULTS

A. Development Environment

B. Methods

C. Models

D. Structured Evidence

E. Ablations

F. Error Analysis

V. DISCUSSION

VI. CONCLUSION

Entity	Claim it supports	Evidence shown	Reviewer risk
Table 1: Dataset Summary	Method is robust and generalisable	Dataset size, labels, splits, class balance	How valuable is synthetic data?
Table 2: Main Results	Method improves over baseline in 3 key ways	F1, evidence, schema validity, uncertainty	What do schema and evidence validity mean?
Figure 1: Architecture	Method is modular, transparent, reproducible, and clinically logical	Pipeline stages, representative example	Architecture may seem unnecessarily convoluted
Table 3: Ablation	Deterministic and LLM components are complementary	LLM-only, rules-only, LLM + schema validation, LLM + evidence verifier, LLM +	Unclear separation between rules and LLM; unclear rules

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Please number citations consecutively within brackets [1]. The sentence punctuation follows the bracket [2].

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